

Unrelated Is Not Opposed: A Signed Scale for Meaning Preservation

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Abstract

Metrics for meaning preservation report how much meaning two sentences share, on a scale from 0 to 100. That scale cannot express opposition. Because contradicting sentences are about the same thing, they overlap heavily, and a released model scores “I drink milk” against “I do not drink milk” at 58 out of 100. We propose a signed scale on $[-100, 100]$, obtained by multiplying a magnitude head by $1 - \alpha p_{\text{contra}}$ from a three-way polarity head. The form is a product so that unrelated sentences score 0 whatever their polarity. Across 140 runs we find that a polarity head reaches 91.1% macro- F_1 , that derived-polarity augmentation leaves task performance unchanged while raising generated sanity suites from 79–90% to 98–99% on all seven encoders, and that composition puts 84% of SICK contradictions below zero where the magnitude alone puts none.

1 Introduction

Assessing whether a rewritten sentence preserves the meaning of its source is a prerequisite for evaluating text simplification, summarisation and translation (Saggion, 2017). The metrics built for it, whether embedding-based (Zhang et al., 2020; Reimers and Gurevych, 2019), question-based (Scialom et al., 2021) or trained end to end (Maddela et al., 2023; Beauchemin et al., 2023), all return a single non-negative number. A score of 0 means the sentences have nothing in common and a score of 100 means the meaning is intact.

However, that output shape has a consequence that is easy to state and hard to see in aggregate correlations. The scale has no room for opposition. Two sentences that contradict each other are, by construction, about the same thing, so they share most of their lexical material and most of their subject matter; a magnitude metric reads that overlap and reports a middling-to-high score. Measured rather than supposed: a released model scores “I drink milk” against “I do not drink milk” at 58.3, and the SICK contradiction “two dogs are wrestling and hugging” against “there is no dog

wrestling and hugging” at 51.7. Both sit on the positive half of a scale on which the positive half is the half that means the meaning survived. Baral (2026) quantified the effect on legal text: across minimal perturbations that reverse legal force while preserving most tokens, embedding metrics and BERTScore move over a small fraction of their range, while bidirectional inference models move substantially. The metrics are not merely imprecise on these pairs; they are confidently wrong in the direction that matters.

We begin by measuring the failure on the metric we build on. The published MeaningBERT model, evaluated on a held-out set of 12 000 class-balanced pairs, ranks agreement above contradiction with an AUC of 60.8%, and on SICK (Marelli et al., 2014) alone it reaches 48.0% with an amplitude that is negative: it scores contradictions marginally higher than agreements. Consequently, a metric in production today tells its users that a negated sentence preserves the meaning of its source.

This paper proposes to make the scale signed, $[-100, 100]$, where -100 denotes opposed meaning and 0 denotes no relation. Our contributions are:

1. **A signed scale obtained by composition, not by direct supervision** (Section 3). We found no corpus carrying human-annotated signed scores, so fitting one would train a model on our own conversion rule. We instead compose two heads whose targets both have abundant human labels, and calibrate the composition on the single corpus that annotates both dimensions over the same pairs.
2. **A measured diagnosis and a 140-run grid** (Section 5, Section 6). Seven encoders, ten seeds, two corpus conditions, with class- and corpus-stratified evaluation and source-sentence leakage removed in both directions.
3. **Three results read off that grid** (Section 6): a polarity head is learnable at 91.1% macro- F_1 ; derived-polarity augmentation buys sanity and not performance, on all seven encoders; and at large

size inference pretraining buys reasoning under negation rather than aggregate accuracy, a separation the base-size pair does not reproduce.

2 Related Work

Meaning preservation metrics. BERTScore (Zhang et al., 2020) computes token-level similarity from contextual embeddings; SARI (Xu et al., 2016) scores edit operations; QuestEval (Scialom et al., 2021) asks questions of both texts; LENS (Maddela et al., 2023) and MeaningBERT (Beauchemin et al., 2023) are trained against human judgements. All of them emit a magnitude, and all of them are evaluated by correlation with human ratings plus, in the case of MeaningBERT, two sanity checks: identical sentences must score 100, unrelated sentences must score 0. Baral (2026) observes that both checks are satisfied by any monotone function of lexical overlap, which is precisely why they do not catch the failure we study.

Inference and its probes. Natural language inference supplies the missing dimension. Large-scale corpora (Williams et al., 2018; Nie et al., 2020) and fact verification (Thorne et al., 2018; Schuster et al., 2021) provide the three-way entailment, neutral and contradiction labels. Targeted suites test what the aggregate hides: Geiger et al. (2020) probes downward monotonicity, where a lexical substitution inside the scope of a negation reverses the direction of entailment; NaNLI (Truong et al., 2022) probes negation that scopes below the clause. PAWS (Zhang et al., 2019) supplies high-overlap non-paraphrases, though we exclude it (Section 4).

Why neither alone. An inference model answers whether the second sentence follows from or denies the first, and says nothing about how much meaning they share; a preservation metric answers the converse. Marelli et al. (2014) is, to our knowledge, the only resource that annotates both over the same pairs, which makes it the only place the composition can be calibrated.

3 A Signed Scale

What the number means. We define the score so that +100 is identical meaning, 0 is no relation, and −100 is opposed meaning. The decisive property is that the negative half measures active contradiction, not the absence of shared meaning. Two sentences about different topics score 0, not −100: to contradict each other, two sentences must first be about the same thing.

Thus, the score carries two pieces of information. The magnitude asks what the sentences talk about together, and is exactly the quantity existing metrics estimate. The polarity asks, once the subject is shared, whether they assert the same thing or its negation.

Alternatives. Two other designs were considered. Typed penalties in the style of MQM, subtracting a cost per error category from a perfect score, give an interpretable and natively negative output, but they require an error taxonomy annotated over the same pairs and they change the target rather than extend it. Learning the signed scale end to end is discussed next and rejected for a reason that applies to any such target.

Why composition rather than direct supervision.

We are not aware of a corpus carrying signed human annotation: we found no resource that asks annotators to rate “how strongly these sentences say the opposite” on a continuous scale. Learning $[-100, 100]$ end to end would require manufacturing the target from categorical labels, that is, training the model on our own conversion rule and then reporting the rule rather than a human judgement. Composition avoids this because both components have abundant human labels of their own, and because each head stays separately falsifiable: when the composed score is wrong, we can say which half moved.

The form. We take

$$s = \text{clip}_{[-100, 100]}(m \times (1 - \alpha p_{\text{contra}})), \quad (1)$$

with magnitude $m \in [0, 100]$ and $p_{\text{contra}} \in [0, 1]$ from the polarity head. A product, not a difference: an unrelated pair has $m \approx 0$ and therefore scores near zero whatever its polarity, whereas a subtraction would let it drift negative and destroy the distinction the scale rests on. Table 1 works three cases.

Why $\alpha = 2$. The slope governs how much contradiction is needed to flip the sign. A certain contradiction on a perfect magnitude lands at $100 \times (1 - \alpha)$, so the negative half of the scale is reachable only from $\alpha = 2$ upwards; below it the scale is truncated. We treat reachability as a constraint on the search rather than as a term in the objective, because it is the definition of the scale and not a quantity to be traded against the others. Section 7 reports what that constraint costs.

Pair	m	p_{contra} (%)	s
“I drink milk” / “I consume milk”	96.7	0.5	+95.8
“I drink milk” / “I do not drink milk”	58.3	99.7	−58.0
“I drink milk” / “the meeting is at three”	0.0	0.5	+0.0
“Two dogs are wrestling” / “there is no dog wrestling”	51.7	99.4	−51.1

Table 1: Measured, not illustrative: m from a released magnitude model, p_{contra} from our polarity head, s from Equation 1 with $\alpha = 2$. Rows two and four are the ones a magnitude scale gets backwards, reporting half the meaning preserved for a negation. Row three shows why the composition is a product: with $m = 0$ the polarity cannot drag an unrelated pair negative.

Corpus	Rows	Labels	Role
VitaminC	488 904	FACT3	train
SICK	9 840	NLI3 + rel.	train, bridge
MoNLI	1 202	ent./neutral	probe
NaN-NLI	258	NLI3	probe

Table 2: Corpora. FACT3 is SUPPORTS, NOT ENOUGH INFO, REFUTES; harmonisation maps it onto NLI3, the same relation under a fact-checking name.

4 Data

Corpora. We merge four corpora that carry inference labels over sentence pairs (Table 2). Schuster et al. (2021) supplies the bulk: its pairs are contrastive and minimal by construction, since a Wikipedia sentence was edited so that the claim its previous revision supported is refuted by the next. SICK contributes the bridge, being the only corpus with both a relatedness score and an inference label over the same pairs. MoNLI and NaN-NLI are held out entirely as probes and never enter training, which the corpus builder enforces rather than documents.

Moreover, we considered and rejected PAWS. Its binary paraphrase label is not a polarity judgement: two sentences that are not paraphrases need not contradict each other, and mapping not a paraphrase onto contradiction would teach the head that difference amounts to opposition, which is the failure the work exists to remove.

Derived-polarity augmentation. Our second corpus condition adds generated pairs. Every added row has its

polarity derived, never copied, which matters because polarity is not symmetric the way magnitude is. If A contradicts B then B contradicts A , so a mirrored contradiction remains one; but “a dog is running” entails “an animal is running” and the converse is false, so a mirrored entailment loses its label rather than carrying a wrong one. We add three families in equal quotas of 25 000: mirrored contradictions, identical pairs labelled entailment, and low-overlap random pairs labelled neutral. The last is the one the signed scale depends on, since it is what teaches that unrelated is not opposed. Equal quotas are necessary: mirroring alone would take the contradiction class from 50 641 to 101 282 against 51 274 entailments and teach the head to over-predict the one class whose sign inverts the score.

Splits. Development and test splits are drawn stratified by class and by corpus, 4 000 per class. Stratification by corpus is not cosmetic: SICK is 8% of the naturally merged test split because VitaminC is fifty times its size, and selecting a model on a split that is almost entirely VitaminC selects for the wrong thing. Stratification raises SICK to 34% of the test split. We then remove source-sentence leakage in both directions, which matters here because VitaminC reuses the same evidence across many claims and splits by case rather than by evidence: 5.6% of the test groups shared a source sentence with training. Appendix A gives the construction and the exact counts.

5 Experimental Setup

We train a three-way classification head on seven encoders $\times$ ten seeds (42–51) $\times$ two corpus conditions, for 140 runs. The conditions are RAW (152 060 training rows) and AUG (227 060). Encoders span three families, BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019) and DeBERTaV3 (He et al., 2023), and two sizes, and include both plain and inference-pretrained variants of the same architecture so the effect of inference pretraining is measured at constant size.

Checkpoints already fine-tuned for three-way inference keep their head when its class order matches ours, which is free and on-task supervision; where the published order is reversed we permute the classifier rows rather than silently training from a permuted head. Every run shares one set of optimisation settings (Appendix B), and the hardware is described in Appendix C.

Moreover, the two halves of the metric cost very different amounts. The grid is 140 training runs and roughly 500 GPU-hours; the composition of Section 7

trains nothing, fits one parameter and runs on CPU in under half an hour. We state the asymmetry here because it is the practical argument for composing two heads rather than training a signed target end to end, and we leave the per-run cost to Appendix C.

We report macro- F_1 on the stratified test split, accuracy on three generated sanity suites built from test-split sentences (identical, unrelated, mirrored), and accuracy on the two held-out probes. Significance uses Welch’s t -test (Welch, 1947) between conditions and a paired test by seed between architectures, with Cohen’s d beside every p -value: with ten seeds and standard deviations near two tenths of a point, a difference of one tenth reaches significance while meaning nothing. The condition comparison is fourteen tests, seven encoders on two axes, so we apply Holm-Bonferroni (Holm, 1979) across the family. It changes no verdict: the ten differences that were significant stay so at the corrected thresholds, and the four that were not remain above them. We report the uncorrected p -values and say this once rather than carrying two numbers through every sentence.

6 Results

What the task is worth against nothing. A three-way score on a balanced split means little on its own, so Table 4 puts three references beside it. The majority class sits at 16.7% macro- F_1 , which is the arithmetic floor. Jaccard overlap between the two token sets, cut by two thresholds fitted on development data, reaches 39.4%, and a TF-IDF bag of unigrams and bigrams with a linear model reaches 45.6%. Consequently, roughly half of the task is surface statistics of the corpus and the fine-tuned head is worth 45 points above the best of the three. The overlap baseline carries a second reading we return to below: ranking entailment above contradiction, it reaches an AUC of 63.5%.

The polarity head is learnable. The best configuration reaches $91.1 \pm 0.14\%$ macro- F_1 on a class-balanced test set of 12 000 pairs (Table 3). The seed standard deviation of 0.14 points over ten runs indicates the result is a property of the configuration and not of a draw.

Augmentation buys sanity, not performance. On the four DeBERTa encoders the macro- F_1 difference between conditions lies between -0.21 and $+0.08$ points and is not significant (p from 0.17 to 0.74); on the other three it is significant but below one point. The sanity suites move in a different regime entirely (Table 5, Fig-

ure 1): $+8.4$ to $+20.0$ points, $p < 10^{-4}$ on all seven encoders, with d from 9.9 to 81.1. The effect sizes are what settle the matter. On RoBERTa-base_{STS-B}, $d = 81$ places the shift two orders of magnitude above the seed-to-seed noise, whereas the largest task effect anywhere in the grid is $d = 2.2$ for a difference of 0.73 points. Every encoder converges to 98.3–98.8% regardless of where it started, and the clearest cases are the weakest models: RoBERTa-base_{STS-B} moves from 78.7% to 98.7% and BERT-base from 79.4% to 98.3%.

Indeed, this is the behaviour the augmentation was built for. It does not make the head better at the task; it repairs how the head behaves at the ends of the scale, which is where a signed metric is read and where the unrelated-as-opposed failure lives.

One caveat belongs here rather than in a footnote. The sanity suites are built by the same three generators that produce the augmented training rows, over different sentences but by the same construction. An AUG model is therefore evaluated partly on a distribution it has seen the shape of, and the gain in Table 6 is not a claim of generalisation to arbitrary degenerate pairs. What the comparison does establish is that the behaviour at the ends of the scale is learnable at all and is not acquired incidentally: seven encoders trained without these rows converge nowhere near 98%, and the two that start lowest are the two that end up moving most.

Which suite moves. However, the aggregate hides where the gain comes from, so Table 6 separates the three families. Identical pairs are essentially solved before augmentation, which is the result one should expect from a trivial case. The gain is concentrated on the other two. Unrelated pairs rise from 92–96% to 99.3–99.7%: before augmentation, between 2.4% and 7.9% of sentence pairs with nothing in common are predicted contradiction, which on the signed scale means returning a negative score where the truth is zero. Mirrored contradictions rise from 74–75% to 96%, and on BERT-base from 50%, meaning that the symmetry of contradiction is not something these encoders hold on their own.

Inference pretraining buys reasoning under negation. Paired by seed, DeBERTa-v3-large_{NLI} beats its plain counterpart by $+0.19$ points of macro- F_1 ($p = 0.002$, $d = 1.39$) and by $+12.1$ points on MoNLI ($p < 10^{-4}$, $d = 2.58$). The aggregate metric barely registers an advantage that the monotonicity probe shows

Encoder	Condition	macro-F ₁	Sanity suites	NaN-NLI	MoNLI
DeBERTa-v3-large _{NLI}	RAW	91.1 ± 0.14	90.4 ± 0.47	72.9 ± 0.65	34.9 ± 3.13
	AUG	91.0 ± 0.31	98.8 ± 0.04	71.9 ± 1.83	31.2 ± 5.11
DeBERTa-v3-large	RAW	90.9 ± 0.10	88.9 ± 0.42	70.6 ± 1.17	22.8 ± 5.22
	AUG	90.7 ± 0.43	98.7 ± 0.40	69.1 ± 1.24	16.0 ± 5.84
RoBERTa-large _{MNLI}	RAW	89.5 ± 0.10	84.6 ± 1.07	68.1 ± 1.02	7.0 ± 1.10
	AUG	89.7 ± 0.10	98.7 ± 0.07	66.6 ± 1.09	8.3 ± 1.68
DeBERTa-v3-base _{NLI}	RAW	89.5 ± 0.15	87.5 ± 0.37	68.6 ± 0.62	11.5 ± 2.39
	AUG	89.6 ± 0.17	98.7 ± 0.08	67.4 ± 0.67	11.6 ± 2.05
DeBERTa-v3-base	RAW	88.8 ± 0.63	86.3 ± 1.74	66.4 ± 1.68	13.4 ± 2.96
	AUG	88.7 ± 0.35	98.6 ± 0.13	64.1 ± 1.92	10.9 ± 3.27
RoBERTa-base _{STS-B}	RAW	85.9 ± 0.15	78.7 ± 0.34	63.8 ± 0.99	4.6 ± 0.60
	AUG	86.2 ± 0.13	98.7 ± 0.08	62.6 ± 0.74	4.9 ± 0.64
BERT-base	RAW	77.7 ± 0.38	79.4 ± 0.65	46.4 ± 2.70	13.6 ± 5.52
	AUG	78.4 ± 0.26	98.3 ± 0.10	49.0 ± 1.13	17.5 ± 2.57

Table 3: Polarity head, mean and standard deviation over ten seeds, in percent. RAW trains on the merged corpus, AUG adds the derived-polarity augmentation. Sanity is accuracy on the three generated suites; NaN-NLI and MoNLI are held-out probes the models never train on. **Bold** marks the best value in each column, higher being better everywhere.

Model	macro-F ₁	AUC
Majority class	16.7	–
Token overlap	39.4	63.5
TF-IDF, logistic regression	45.6	68.5
Best fine-tuned head	91.1	–

Table 4: Baselines on the same test split, in percent. The majority class is the arithmetic floor of a split balanced at 4 000 per class, where it also scores 33.3 accuracy. Token overlap is Jaccard over the two token sets, cut by two thresholds fitted on development data. AUC ranks entailment above contradiction and leaves neutral pairs out. **Bold** marks the best macro-F₁, which is the fine-tuned head of Table 3.

plainly. At base size the pattern inverts: the task gain is larger (+0.67 points, $p = 0.005$) and the MoNLI difference is not significant. We read this as inference pretraining supplying a capability that the aggregate test set does not exercise, which is the argument for reporting probes alongside it.

How the models fail on monotonicity. MoNLI accuracy is low for every configuration, and the confusion matrix shows the failure is not the one we expected. The heads almost never call these pairs contradictions: on DeBERTa-v3-large_{NLI}, 0.2% of true entailments are predicted contradiction. They have not learned that the

token “not” signals opposition. What they fail is the reversal itself: 93% of true entailments are predicted neutral, because under negation the more specific term follows from the more general one rather than the other way round, and the heads apply the unreversed rule. This is a clean negative result, and it is what the probe exists to expose: no aggregate number on our test split distinguishes a head that has learned the relation from one that has learned its unnegated half.

Diagnosis of deployed metrics. Table 7 scores three released magnitude models on both tasks at once: Pearson against human meaning-preservation ratings, and the ability to rank agreement above contradiction on our test split. The two columns are close to independent. The model with the weakest correlation separates polarity better than the one with the strongest, and the BERT-base variant, which is a competent magnitude estimator at $r = 0.640$, sits near chance at an AUC of 53.7%. On SICK alone the published model reaches 48.0% with a negative amplitude: it scores contradictions above agreements. The last row of Table 7 is the one we find hardest to look away from. Thresholded token overlap, which contains no model, ranks entailment above contradiction at 63.5%, above the published metric’s 60.8% and above the BERT-base variant’s 53.7%. A metric trained on human meaning-preservation judgements therefore orders polarity no better than counting shared words, which is the claim of

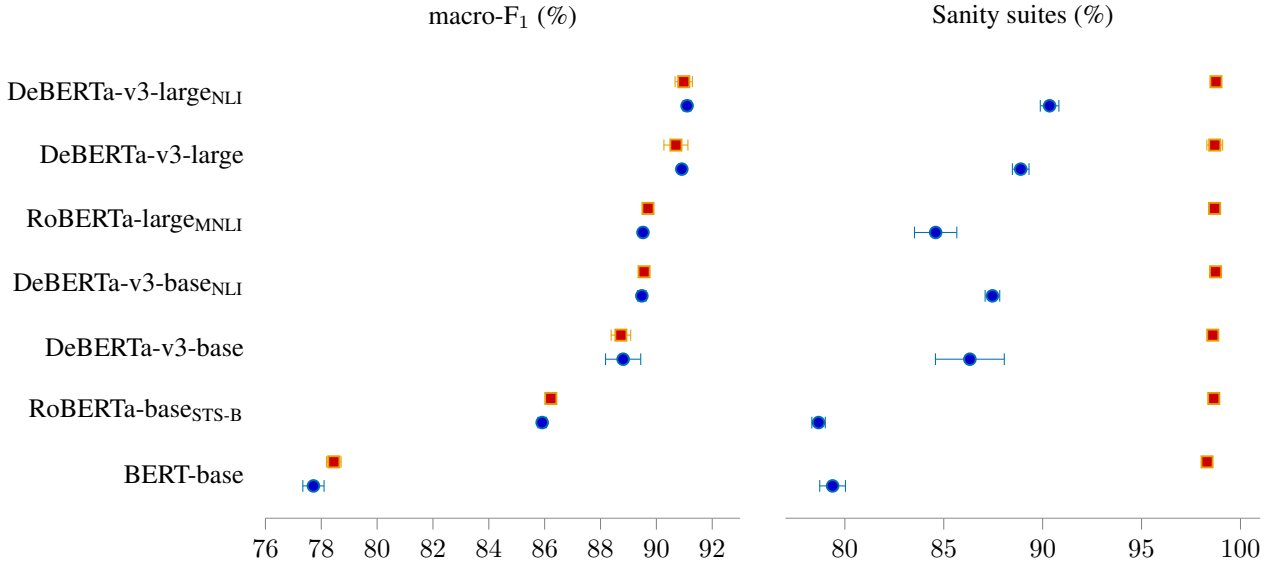


Figure 1: Augmentation moves one axis and not the other. Blue circles are the RAW condition, without augmentation; orange squares the AUG condition, with it. Points are means over ten seeds and bars one standard deviation. On the task axis the two conditions overlap for every encoder; on the sanity axis they converge to one value from seven different starting points. Note the two panels do not share an x range.

Encoder	macro-F ₁		Sanity suites	
	Δ (pp)	d	Δ (pp)	d
DeBERTa-v3-large _{NLI}	$-0.11 \pm 0.11^\dagger$	-0.47	$+8.40 \pm 0.15$	+25.4
DeBERTa-v3-large	$-0.21 \pm 0.14^\dagger$	-0.67	$+9.81 \pm 0.18$	+24.1
RoBERTa-large _{MNLI}	$+0.19 \pm 0.05$	+1.79	$+14.10 \pm 0.34$	+18.6
DeBERTa-v3-base _{NLI}	$+0.08 \pm 0.07^\dagger$	+0.49	$+11.28 \pm 0.12$	+42.2
DeBERTa-v3-base	$-0.08 \pm 0.23^\dagger$	-0.15	$+12.28 \pm 0.55$	+9.9
RoBERTa-base _{STS-B}	$+0.31 \pm 0.06$	+2.22	$+19.98 \pm 0.11$	+81.1
BERT-base	$+0.73 \pm 0.15$	+2.22	$+18.94 \pm 0.21$	+40.9

Table 5: Effect of augmentation in percentage points, AUG minus RAW, with the standard error of the difference and Cohen’s d , from Welch’s t -test over ten seeds. $\dagger$ marks a difference that is not significant at $p < 0.05$. Every sanity difference has $p < 10^{-4}$. The two axes are reported together on purpose: the claim is that one moves and the other does not.

Baral (2026) measured on our split rather than argued.

Two rows at the bottom answer the question this invites. An off-the-shelf inference model, used as published with no training of ours, ranks entailment above contradiction on our test split at an AUC of 96.6%, well above every magnitude model. Its three-way accuracy is another matter: 68.4% macro-F₁ against 91.1% for the same encoder fine-tuned on our corpus. The distinction is worth stating plainly, because it bounds what this paper contributes. For the ranking question alone, which is all the composition consumes through p_{contra} , a released inference model is already strong and fine-tuning adds little; the +22.7 points of macro-F₁ that fine-tuning buys is calibration of three classes onto this domain, not the discovery of the relation. A

practitioner who wants only the sign, and is willing to fit α against a miscalibrated probability, may not need to train anything at all. One asymmetry in this comparison is ours to declare: we report the off-the-shelf model’s AUC and our head’s macro-F₁, because the grid stores predicted classes and not the probabilities an AUC needs. The ranking comparison is therefore one-sided, and whether fine-tuning moves the AUC at all is open.

Finally, the exception among the magnitude models is instructive. DeBERTa-v3-large reaches an AUC of 81.5% without ever seeing a polarity label, and the BERT-base model trained on the same corpus with the same recipe reaches 53.7%. The separation therefore comes from the encoder’s pretraining, not from

Encoder	Condition	Identical	Unrelated	Mirrored
DeBERTa-v3-large _{NLI}	RAW	100.0 $\pm$ 0.00	95.9 $\pm$ 0.78	75.1 $\pm$ 1.20
	AUG	100.0 $\pm$ 0.00	99.7 $\pm$ 0.07	96.5 $\pm$ 0.13
DeBERTa-v3-large	RAW	100.0 $\pm$ 0.00	92.7 $\pm$ 0.60	73.9 $\pm$ 1.46
	AUG	100.0 $\pm$ 0.04	99.7 $\pm$ 0.13	96.4 $\pm$ 1.14
BERT-base	RAW	96.5 $\pm$ 0.16	92.0 $\pm$ 0.67	49.7 $\pm$ 1.74
	AUG	100.0 $\pm$ 0.00	99.3 $\pm$ 0.09	95.6 $\pm$ 0.29

Table 6: The three generated suites separately, mean and standard deviation over ten seeds, in percent. Identical pairs must be entailment, unrelated pairs neutral, mirrored contradictions still contradiction. The unrelated column is the one the signed scale depends on, since it is where a pair with nothing in common is read as opposed.

Magnitude model	Preservation r	Polarity AUC (%)
MeaningBERT (published)	0.323	60.8
BERT-base (ours)	0.640	53.7
DeBERTa-v3-large (ours)	0.705	81.5
<i>Off-the-shelf inference model, no fine-tuning</i>		
DeBERTa-v3-large _{MNLI}	–	96.6
DeBERTa-v3-base _{MNLI}	–	96.0
<i>Surface baseline</i>		
Token overlap, Jaccard	–	63.5

Table 7: Both tasks, one test split. Pearson is on the 1 536 human-annotated pairs of the preservation test set, excluding its 116 generated controls, which inflate the figure by up to +0.21. AUC ranks entailment above contradiction on our 12 000-pair split, in percent. The inference models produce no preservation score, hence the dashes. **Bold** marks the best polarity ranking in each of the two trained blocks. The last row is a thresholded token overlap, carried here so the magnitude models are read against something that has no model in it at all.

the preservation data. Nothing in a magnitude metric’s training signal distinguishes a contradiction from a failed paraphrase, since both receive the same low human rating, so no quantity of meaning-preservation data can produce polarity. That is the argument for a second head rather than a better-trained first one.

7 Composing the Two Heads

We fit α on the SICK half of the development split and report on the SICK half of the test split, 4 116 pairs of which 712 are contradictions. Fitting on test and reporting on test would be tuning on the test set, even for one parameter over a coarse grid. The components are a released magnitude model and one of our polarity heads, DeBERTa-v3-large trained on the RAW condition.

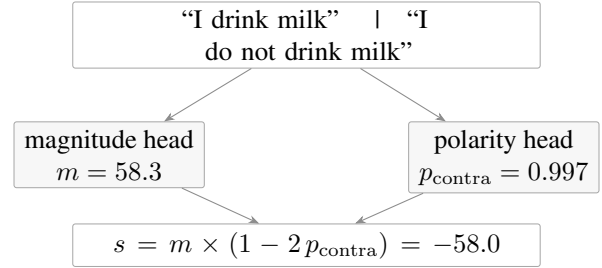


Figure 2: One pair, end to end. The two heads are read independently and the composition of Equation 1 turns a middling positive score into a negative one. The same pipeline on “I drink milk” against “I consume milk” gives $m = 96.7$, $p_{\text{contra}} = 0.005$ and $s = +95.8$, and on an unrelated pair $m = 0.0$ sends the product to zero whatever the polarity says.

Because no signed gold score exists, we select α against three requirements multiplied together, so that sacrificing one cannot hide behind the others: contradictions must fall below zero, entailments must stay above it, and on the pairs whose sign does not flip the score must still track human relatedness. Furthermore, a composition that sends every contradiction to -100 and everything else to noise would satisfy the first two and is excluded by the third. The development split selects $\alpha = 2$, the same value the reachability constraint of Section 3 forces, so the two arguments agree rather than compete.

Table 8 gives the result. Indeed, the magnitude alone puts no contradiction at all below the line, which is not a weakness of that particular model but a property of a non-negative scale. Composition puts 84.4% of them where they belong while retaining $r = 0.760$ with human relatedness on the unflipped pairs. The cost of requiring the full scale is visible and small: $\alpha = 1.25$ scores 2.3% higher on the combined objective and truncates the negative half at -25 .

Slope	Contradictions below zero (%)	Entailments above zero (%)	Relatedness r	Reachable floor
$\alpha = 1.00$	0.0	100.0	0.820	0
$\alpha = 1.25$	81.2	99.9	0.807	-25
$\alpha = 1.50$	82.6	99.8	0.792	-50
$\alpha = 1.75$	83.6	99.8	0.777	-75
$\alpha = 2.00$	84.4	99.8	0.760	-100
$\alpha = 3.00$	86.2	99.7	0.689	-100
Magnitude alone	0.0	100.0	0.852	0

Table 8: Sensitivity of the composition on the SICK half of the test split. The slope itself was fitted on development data, which selects $\alpha = 2$. Slopes below 2 cannot reach -100 and are excluded by construction; among all slopes the unconstrained optimum on test would be $\alpha = 1.25$, better by 2.3% relative and capped at -25. **Bold** marks the slope the paper adopts, not the best value in any column.

Polarity head	Off-the-shelf	Fine-tuned
Contradictions below zero	97.6	84.4
Relatedness r	0.599	0.760
Combined objective	0.574	0.640

Table 9: The same magnitude model composed with two polarity heads at $\alpha = 2$, on the SICK half of the test split. The first row is a percentage of the 712 contradictions. The off-the-shelf head is DeBERTa-v3-large_{MNLI} used as published; the fine-tuned one is ours. **Bold** marks the better value of each row, higher being better on all three.

What fine-tuning buys the composition. The off-the-shelf inference model ranks polarity at an AUC of 96.6% without training, which invites the question of whether our head earns its cost. Table 9 composes each head with the same magnitude model, under the same protocol. The off-the-shelf head is the better sign detector: at $\alpha = 2$ it puts 97.6% of contradictions below zero against 84.4%. However, it costs 0.161 of correlation with human relatedness on the pairs whose sign does not flip, 0.599 against 0.760, and loses the combined objective by 11% relative. Consequently, what fine-tuning buys is not the relation but its calibration: an uncalibrated head emits a confident p_{contra} on pairs that are merely unrelated, and the product then drags their magnitude down. This is the third requirement doing exactly the work it was introduced for, and it is why we select α against a product rather than against the sign criterion alone.

8 Limitations

While we believe the signed scale is a useful contribution to the evaluation of meaning preservation, we acknowledge six limitations that bound our claims and

offer avenues for future research.

Augmentation costs monotonicity. The gain in sanity is paid for on MoNLI, which drops on three of seven encoders and on both large DeBERTa variants, by -6.8 points on the plain one and -3.7 on the inference-pretrained one. Generated identical and unrelated pairs teach the easy ends of the scale, not the reversal of entailment under negation. We report the trade rather than resolve it.

The ranking comparison is one-sided. We report the off-the-shelf model’s AUC and our head’s macro- F_1 , because the grid stores predicted classes rather than the probabilities an AUC needs. Table 9 answers the practical half of the question, since it composes both heads and measures the result, but it leaves the ranking itself unmeasured on our side. Whether fine-tuning improves the ordering of p_{contra} , rather than only its calibration, is the one quantity this paper still owes its reader.

No signed gold standard. The composition is calibrated against three necessary conditions, not against human signed judgements, because none exist. A corpus of human-rated signed pairs would let the form of Equation 1 be tested rather than argued, and we regard collecting one as the natural next step.

One corpus carries the test set. Stratification raises SICK to a third of the test split, which leaves VitaminC with the other two thirds. The headline macro- F_1 is therefore mostly a VitaminC number, and VitaminC pairs are claim-evidence pairs from Wikipedia revisions rather than the sentence pairs a simplification

metric usually sees. We did not measure the head separately on each corpus, and a per-corpus breakdown of Table 3 would say how much of the result transfers.

The bridge is thin on one side. SICK supplies only 495 development rows, of which 229 shared source sentences with its own test portion and were removed, leaving the development split at 2.3% SICK against 34% in test. Model selection is therefore driven by VitaminC while the test set is not. This is a property of the available data, not a design choice, and it is the first thing to watch should the two numbers diverge.

Scope. All experiments are English, sentence-level, and run in fp32 on Pascal GPUs (Appendix C); we did not evaluate encoders requiring more recent hardware.

9 Conclusion and Future Works

A meaning-preservation metric that cannot distinguish a paraphrase from a negation is not imprecise, it is confidently wrong on the pairs where precision matters, and we show that the metric deployed today has exactly this property. Making the scale signed does not require a better magnitude model; it requires a second head and an explicit rule for combining them. The combination is cheap, since it trains nothing beyond a three-way classifier and fits one parameter, and it moves 84% of contradictions to the correct side of zero where the magnitude alone moves none. What it does not yet have is a human-annotated signed target against which the combining rule itself could be tested.

Ethical Considerations

A meaning-preservation metric is deployed to decide whether a rewritten sentence may be shown to a reader in place of the original. When such a metric reports that a negated sentence preserves the meaning of its source, as we measure released models doing, the error reaches the reader rather than the engineer. Indeed, the risk is concrete in the settings where simplification is most useful, where a layperson may rely on the rewritten text without access to the original (Beauchemin et al., 2023). However, a signed scale does not remove that risk, but it makes the failure visible to the system that consumes the score instead of hiding it inside a positive number. Furthermore, our corpora inherit the biases of the Wikipedia revisions and the caption corpora they are drawn from, and a polarity head trained on them will carry those biases into whatever it gates (Bender et al., 2021).

Availability

The corpus builder, the loaders, the training and composition code, the 140 result files and the script that generates every table and figure in this paper will be released under an open licence on acceptance. The merged corpus inherits the most restrictive licence among its sources and will be redistributed under CC BY-NC-SA 4.0. Finally, the loaders are released separately so the corpus can be rebuilt from the original sources.

Use of Large Language Models

This work utilized Large Language Models (specifically Claude) to assist with the linguistic refinement, typographical errors, and LaTeX formatting of the manuscript (table generation, figure improvements). The authors retained full control over the scientific content, data analysis, and conclusions.

Acknowledgements

Removed for double-blind review.

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A Corpus Construction and Leakage Control

Merging and harmonisation. The four corpora are merged into one table with a unified three-way polarity column. VitaminC carries FACT3 labels, which we map onto NLI3 as SUPPORTS to entailment, NOT ENOUGH INFO to neutral and REFUTES to contradiction. SICK is assembled from two public halves, one carrying the inference label and one the relatedness score; the join is verified to lose no more than 1% of rows, and in practice loses none.

Stratification. Development and test splits take 4 000 rows per class, drawn in equal share from each corpus by water-filling: a corpus smaller than its share is taken whole and the shortfall is redistributed over the larger ones. Without it, SICK falls to 8% of the test split and 0.8% of the development split.

Leakage. We group rows by source sentence and move whole groups, never rows. VitaminC reuses one piece of evidence across many claims, so 1 662 source sentences, 5.6% of the test groups, were also present in training. We drop 2 379 training rows and 229 development rows, and assert that train/dev, train/test and dev/test share no pair and no source sentence. The assertion runs again after augmentation, since a generated pair inherits its source sentences.

Augmentation quotas. The three generated families enter in equal quotas of 25 000, which takes the training set from 152 060 rows to 227 060. Mirroring alone would take the contradiction class from 50 641 to 101 282 against 51 274 entailments.

B Training Details

Optimisation. Every run uses two epochs, learning rate 10^{-5} , effective batch 32 obtained by gradient accumulation, and a maximum sequence length of 256 tokens, in fp32. Nothing is tuned per encoder: one setting across the grid is what makes the seven encoders comparable, and the cost is that no encoder is at its own optimum.

Head reuse. Four of the seven checkpoints ship a three-way inference head. We keep it when its class order matches ours and permute its rows when the published order is reversed, as it is for `roberta-large-mnli`. The alternative, training from a permuted head, is silent and costs an epoch of relearning the permutation.

Seeds. Seeds 42 to 51 control initialisation of the new head, data order and dropout. Every number reported in this paper is a mean over the ten, with its standard deviation.

C Hardware and Compute Budget

Hardware. The grid runs on three NVIDIA Pascal-generation cards, a GTX 1080 Ti and a Quadro P5000 on one machine and a Titan Xp on another, in fp32 since the generation has no usable bfloat16. Two of the seven encoders wedge the GTX 1080 Ti reproducibly, which we handle by excluding those two from that card rather than by diagnosing the driver fault.

Budget. The 140 runs take roughly 500 GPU-hours in total. A base-encoder run takes 68 minutes on the RAW condition and 109 minutes on AUG; a large-encoder run takes 318 and 489 minutes. The composi-

tion of Section 7 trains nothing: it is two forward passes over 4 116 pairs plus a grid search over one scalar, and runs on CPU in under half an hour. No private API was used, and the total monetary cost is the electricity of hardware we already own.